# Tianhao Wu

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## Interests

Formal Methods, Control, Machine Learning, Safe Autonomy

## Education

**University of Southern California**
M.S. in Computer Science — Aug 2025 – Present

B.S. in Computer Science and Computer Engineering — Jan 2021 – Dec 2024
B.S. in Applied and Computational Mathematics
GPA: 3.95/4.0

## Publications

**An Algebraic Framework for Quantitative Semantics of Spatio-Temporal Logic with Graph Operators**
Sheryl Paul*, Vidisha Kudalkar*, Anand Balakrishnan*, **Tianhao Wu**, Lars Lindemann, Jyotirmoy V. Deshmukh
*QEST + FORMATS 2026* **[pdf]**

## Preprints

**Scenario MPC with STL Specifications and Pareto-Based Feasibility Repair**
**Tianhao Wu**, Yiwei Lyu
*Under review at IEEE ICRA 2027* **[pdf]** **[project]**

**Enhancing the Performance of DeepReach on High-Dimensional Systems through Optimizing Activation Functions**
Qian Wang*, **Tianhao Wu***
*arXiv 2023* **[pdf]**

## Research Experience

**Research Assistant, University of Southern California** — Apr 2026 – Present
Advisor: Jyotirmoy V. Deshmukh
Implemented algebraic quantitative semantics for Spatio-Temporal Logic with Graph Operators (STL-GO) and evaluated on 2D and 3D multi-agent environments. Investigating STL-GO specifications and their quantitative robustness for credit assignment in cooperative multi-agent reinforcement learning (MARL).

**Research Assistant, Texas A&M University** — Oct 2025 – Present
Advisor: Yiwei Lyu
Developed a model predictive control (MPC) framework with Signal Temporal Logic (STL) specifications under stochastic and uncontrollable agents. Formulated feasibility repair as a Pareto optimization problem and derived a probabilistic certificate for STL violation rate. Evaluated the framework on autonomous-driving scenarios in CARLA with demonstrated safe behaviors.

**Research Assistant, Carnegie Mellon University** Jan 2025 – Apr 2025
Advisor: John M. Dolan
Designed a modular safe-control framework for autonomous driving. Applied Linear Temporal Logic (LTL) specifications for dynamic activation of control barrier function (CBF) constraints. Implemented preliminary safety modules in CARLA.

**Research Assistant, University of Southern California** Aug 2022 – May 2023
Advisor: Somil Bansal
Investigated the performance of DeepReach on high-dimensional Hamilton–Jacobi (HJ) reachability problems with alternative activations. Reduced violation rates by 5.1% on a 9D multi-vehicle collision-avoidance problem.

**Research Assistant, University of California, Irvine** Jun 2022 - Aug 2022
Advisor: Marco Levorato
Built an autonomous drone with obstacle detection and avoidance capabilities. Led end-to-end development, including dataset collection, vision model training, sensor integration, simulation, and field testing.

## Honors & Awards

**Ming Hsieh Institute Undergraduate Scholar** 2023 – 2024
One of the five scholars selected for research potential
**USC Academic Achievement Award** Fall 2023
**Viterbi CURVE Research Fellowship** 2022 – 2023
**Lenore B. Kreiger Scholarship for Mathematics** 2022 – 2023

## Relevant Coursework

**PhD-level:** Robot Learning; Automata Theory, Logic and Formal Verification
**Graduate-level:** Applied Probability; Autonomous Cyber-Physical Systems; Learning and Control for Safety-Critical Robotic Systems

## Teaching Experience

**Teaching Assistant, University of Southern California**

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|---|---|
| Introduction to Artificial Intelligence (CS360) | Spring 2023, Fall 2026 |
| Discrete Methods in Computer Science (CS170) | Fall 2026 |
| Fundamentals of Computation (CS102), Head TA | Spring 2022 – Spring 2024 |
| Introduction to Embedded Systems (EE109) | Fall 2023 |
| CS@SC Summer Coding Camp | Summer 2022 |

## Service

**Reviewer**, Learning and Formal Methods for Robotics, IROS Workshop (2026)

## Skills

**Programming:** Python, C++, MATLAB
**Tools:** Gurobi, CVXPY, PyTorch, CARLA, Linux, Git, LaTeX